

Mock Exam 1 — Solutions

Lectures 1–4 (Chapters 1–3)

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Quantitative Research Methods · HSBI

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Overview of the mock exam

Problem	Topic	Points
P1	Fundamentals of statistical learning	15
P2	Bias–variance decomposition	15
P3	K -nearest neighbours classification	12
P4	Simple linear regression by hand	24
P5	Multiple regression — interpreting Python output	24
Σ		90

Exam conditions

90 minutes for 90 points (≈ 1 point per minute). Allowed aids: non-programmable calculator and one handwritten A4 sheet (both sides). Justify every answer; round final results to 3 decimals.

Problem 1 — task

Problem 1 (15 P) — Fundamentals of statistical learning

- (a) [4 P] Prediction vs. inference — explain the difference; one example each.
- (b) [4 P] Parametric vs. non-parametric estimation of f ; one advantage of each; which flexibility–interpretability trade-off is behind the choice?
- (c) [3 P] Classify: (i) churn prediction for 800 telecom customers; (ii) which firm characteristics are associated with CEO salary (90 firms); (iii) find groups among 5,000 shopping baskets.
- (d) [4 P] Why is training MSE generally smaller than test MSE? What is overfitting?

Problem 1 — solution (a): prediction vs. inference

Solution P1(a) — prediction vs. inference (4 P)

We estimate f in $Y = f(X) + \varepsilon$; the *goal* decides what we need from $\hat{f}$.

Prediction. We only want an *accurate output* $\hat{Y} = \hat{f}(X)$ for new inputs X . The form of $\hat{f}$ is secondary — it may be treated as a *black box*, because the accuracy of $\hat{Y}$ is all that matters.

Example: Is an incoming e-mail spam or not? (Or: predict next month's product demand.)

Inference. We want to understand *how* Y changes with $X_1, \dots, X_p$: *which* predictors matter, in *which* *direction*, and *how strongly*. Here $\hat{f}$ must be *interpretable*.

Example: Which advertising channels increase sales, and by how much per euro spent?

In one line

Prediction targets the *output* $\hat{Y}$; inference targets the *mechanism* inside $\hat{f}$.

Problem 1 — solution (b): parametric vs. non-parametric

Solution P1(b) — two ways to estimate f (4 P)

Parametric — a two-step approach.

- (1) Assume a functional form for f , e.g. the linear model $f(X) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$.
- (2) Estimate the finite set of parameters (e.g. by least squares).

Estimating a whole function is thus reduced to estimating a few numbers. *Advantage:* needs relatively few observations; simple to fit and easy to interpret.

Non-parametric. No explicit form is assumed; $\hat{f}$ is built to follow the data directly (e.g. K -nearest neighbours, splines). *Advantage:* captures a far wider range of shapes of f — no risk of imposing a badly wrong form.

Flexibility–interpretability trade-off. More flexible methods can reduce bias, but need many observations, risk overfitting and are harder to interpret; when *inference* is the goal we usually prefer the more restrictive, interpretable models.

Problem 1 — solution (c): classifying the scenarios

Solution P1(c) — regression / classification / unsupervised (3 P)

- (i) **Classification** (supervised). The response “cancels: yes/no” is *qualitative*; the goal is prediction.
- (ii) **Regression** (supervised). The response CEO salary is *quantitative*; the goal is inference — which factors are associated with salary, and how strongly.
- (iii) **Unsupervised learning**. There is *no response variable*; the goal is to find clusters of similar customers.

Decision rule

Is there a response? *Qualitative* → classification; *quantitative* → regression; *none* → unsupervised.

Problem 1 — solution (d): training vs. test MSE, overfitting

Solution P1(d) — why training MSE < test MSE (4 P)

Why the training MSE is smaller. The method is fitted by making the *training* error small, so $\hat{f}$ adapts to the training observations — *including their random noise*. The training MSE therefore *systematically underestimates* the error on fresh data, and for a given data set training MSE $\leq$ test MSE in general.

Behaviour vs. flexibility. As flexibility grows, the training MSE decreases *monotonically*, while the test MSE first decreases and then increases (U-shape).

Overfitting. The method follows patterns in the *training noise* that do not exist in the population; a *less* flexible method would then achieve a *smaller test* MSE. Symptom: a widening gap — very small training MSE together with a rising test MSE.

Problem 2 — task

Problem 2 (15 P) — Bias–variance decomposition

At a fixed x_0 three methods were analysed; $\sigma^2 = \text{Var}(\varepsilon) = 1.00$:

	A (linear)	B (moderate)	C (very flexible)
$[\text{Bias}]^2$	2.25	0.16	0.01
$\text{Var}(\hat{f}(x_0))$	0.25	1.44	3.60

(a) **[4 P]** State the decomposition of the expected test MSE and explain the components. (b) **[5 P]** Compute the expected test MSE for A–C; which method to choose? (c) **[3 P]** Smallest expected test MSE any method could achieve here? (d) **[3 P]** Behaviour of bias²/variance/training/test MSE as flexibility grows; U-shape (recall test MSEs 3.50, 2.60, 4.61 from (b)).

Problem 2 — solution (a): the decomposition

Solution P2(a) — bias–variance decomposition (4 P)

At a fixed test point x_0 the expected test MSE splits into three non-negative parts:

$$E[(y_0 - \hat{f}(x_0))^2] = \underbrace{\text{Var}(\hat{f}(x_0))}_{\text{sensitivity to training set}} + \underbrace{[\text{Bias}(\hat{f}(x_0))]^2}_{\text{systematic model error}} + \underbrace{\text{Var}(\varepsilon)}_{\text{irreducible noise}} .$$

- **Variance** $\text{Var}(\hat{f}(x_0))$: how much $\hat{f}(x_0)$ would change if the method were re-fitted on a *different* training set.
- **Squared bias** $[\text{Bias}(\hat{f}(x_0))]^2$: the systematic error from approximating the true f by a (possibly too simple) model.
- **Irreducible error** $\text{Var}(\varepsilon) = \sigma^2$: the noise variance of ε , which *no* method can remove.

Problem 2 — solution (b): expected test MSE for A–C

Solution P2(b) — compute the test MSE and choose (5 P)

Expected test MSE = $\text{Bias}^2 + \text{Var} + \sigma^2$, with $\sigma^2 = 1.00$. Adding the three components row by row:

	Bias ²	Var	σ^2	Test MSE
A (linear)	2.25	0.25	1.00	$2.25 + 0.25 + 1.00 = 3.50$
B (moderate)	0.16	1.44	1.00	$0.16 + 1.44 + 1.00 = \mathbf{2.60}$
C (very flexible)	0.01	3.60	1.00	$0.01 + 3.60 + 1.00 = 4.61$

Choose Method B — it has the smallest expected test MSE. Its moderate flexibility buys a large bias reduction ($2.25 \rightarrow 0.16$) at an acceptable variance cost, whereas the very flexible Method C pays more in variance (+2.16) than it gains in bias (−0.15).

Problem 2 — solution (c): the lower bound

Solution P2(c) — smallest achievable test MSE (3 P)

Squared bias and variance are both *non-negative*, so for *every* method

$$E[(y_0 - \hat{f}(x_0))^2] = \underbrace{[\text{Bias}(\hat{f}(x_0))]^2}_{\geq 0} + \underbrace{\text{Var}(\hat{f}(x_0))}_{\geq 0} + \sigma^2 \geq \sigma^2 = 1.00.$$

The floor 1.00 is attained only in the ideal case of *zero bias and zero variance*. Even the true f cannot predict the fresh noise draw ε_0 at x_0 — this $\text{Var}(\varepsilon) = \sigma^2$ is the **irreducible error**.

Interpretation

No modelling effort can push the expected test MSE below $\sigma^2 = 1.00$.

Problem 2 — solution (d): behaviour and the U-shape

Solution P2(d) — flexibility and the U-shape (3 P)

As the flexibility of a method increases:

- **Squared bias** decreases (typically monotonically) — a richer model approximates the true f better;
- **Variance** increases — the fit reacts more to the particular training set;
- **Training MSE** decreases *monotonically* — a more flexible fit follows the training data ever more closely.

The **expected test MSE** $= \text{Bias}^2 + \text{Var} + \sigma^2$ therefore *first falls* (bias reduction dominates) and *then rises* (variance increase dominates) — the characteristic **U-shape**.

In the table the test MSE runs $3.50 \rightarrow 2.60 \rightarrow 4.61$ for $A \rightarrow B \rightarrow C$: the minimum is at the *intermediate* flexibility (B).

Problem 3 — task

Problem 3 (12 P) — KNN classification

Seven visitors with scores X_1 (pages) and X_2 (minutes) and outcome Y (purchase); new visitor at $x_0 = (4,2)$ — classify with Euclidean distance.

Obs.	1	2	3	4	5	6	7
X_1	4	2	6	4	1	7	5
X_2	3	2	2	5	1	4	6
Y	Yes	No	No	No	Yes	Yes	No

- (a) [4 P] All distances; classification with $K = 1$. (b) [4 P] Classification with $K = 3$ and $\hat{Pr}(\text{Yes} \mid x_0)$.
(c) [2 P] Effect of K on bias/variance; training error of 1-NN. (d) [2 P] Bayes classifier: definition; why unattainable; relation to KNN.

Problem 3 — solution (a): distances and $K = 1$

Solution P3(a) — Euclidean distances and $K = 1$ (4 P)

Distance from $x_0 = (4, 2)$: $d(x_0, x_i) = \sqrt{(X_{1i} - 4)^2 + (X_{2i} - 2)^2}$. Two worked cases:

$$\text{Obs. 1: } \sqrt{(4 - 4)^2 + (3 - 2)^2} = \sqrt{1} = 1.000, \quad \text{Obs. 5: } \sqrt{(1 - 4)^2 + (1 - 2)^2} = \sqrt{10} = 3.162.$$

Ranking all seven by squared distance d^2 (order-preserving, avoids square roots):

Obs.	1	2	3	4	5	6	7
d^2	1	4	4	9	10	13	17
d	1.000	2.000	2.000	3.000	3.162	3.606	4.123
Y	Yes	No	No	No	Yes	Yes	No

$K = 1$: the single nearest neighbour is Obs. 1 ($d = 1.000$, label Yes) $\Rightarrow$ classify the new visitor as **Yes**.

Problem 3 — solution (b): $K = 3$ and the probability

Solution P3(b) — majority vote with $K = 3$ (4 P)

KNN estimates each class probability by its frequency among the K nearest neighbours:

$$\hat{\Pr}(Y = j \mid X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} 1(y_i = j), \quad \mathcal{N}_0 = \{K \text{ nearest neighbours}\}.$$

From the ranking in part (a) the three nearest are Obs. 1 ($d = 1.000$, Yes), Obs. 2 ($d = 2.000$, No) and Obs. 3 ($d = 2.000$, No). Hence

$$\hat{\Pr}(\text{Yes} \mid x_0) = \frac{1}{3} \approx 0.333, \quad \hat{\Pr}(\text{No} \mid x_0) = \frac{2}{3} \approx 0.667.$$

Majority vote $\Rightarrow$ classify as **No**. Note that $K = 1$ (Yes) and $K = 3$ (No) *disagree* — the decision depends on the flexibility of the classifier.

Problem 3 — solution (c): the role of K

Solution P3(c) — bias/variance and 1-NN training error (2 P)

Small K (e.g. $K = 1$): very flexible, jagged decision boundary $\Rightarrow$ *low bias, high variance*.

Large K : averaging over many neighbours gives a smooth, almost linear boundary $\Rightarrow$ *high bias, low variance*.

Training error of 1-NN = 0 (ignoring ties): each training point is its *own* nearest neighbour, so it is always classified correctly on the training set — which says *nothing* about its test error.

K is the flexibility knob

Small K = flexible / high variance; large K = rigid / high bias — the same bias–variance trade-off as everywhere.

Problem 3 — solution (d): the Bayes classifier

Solution P3(d) — Bayes classifier and its link to KNN (2 P)

Definition. The Bayes classifier assigns each x to the class j with the largest *true* conditional probability $\Pr(Y = j \mid X = x)$. It minimises the expected test error rate; its error — the **Bayes error rate** — is the lower bound for *all* classifiers.

Why it cannot be used directly. In practice the true conditional distribution of Y given X is *unknown*, so $\Pr(Y = j \mid X = x)$ cannot be evaluated.

Relation to KNN. KNN *mimics* the Bayes rule: it *estimates* the conditional probabilities locally from the K nearest training observations (the part (b) formula) and then classifies to the largest estimate — a data-driven approximation of the Bayes classifier.

Problem 4 — task

Problem 4 (24 P) — Simple linear regression by hand

Radio spots x and weekly sales y (in 1,000 EUR) for $n = 5$ weeks; $\sum x_i = 15$; $\sum y_i = 55$; $\sum x_i^2 = 55$; $\sum x_i y_i = 195$.

Week	1	2	3	4	5
x_i	1	2	3	4	5
y_i	6	7	11	13	18

(a) **[6 P]** $\hat{\beta}_1$ and $\hat{\beta}_0$; fitted line; interpretation. (b) **[4 P]** With $\hat{y} = 2 + 3x$: fitted values and residuals; check $\sum e_i = 0$. (c) **[3 P]** With residuals $(1, -1, 0, -1, 1)$: RSS and RSE; interpretation. (d) **[6 P]** $SE(\hat{\beta}_1)$ (use $\hat{\beta}_1 = 3$, $RSE = 1.155$, $S_{xx} = 10$); t -test of $H_0: \beta_1 = 0$ ($t_{0.975,3} = 3.182$); 95 % CI. (e) **[3 P]** TSS and R^2 (RSS = 4); interpretation. (f) **[2 P]** With $\hat{y} = 2 + 3x$: prediction at $x = 6$; one caveat.

Problem 4 — solution (a): least-squares estimates

Solution P4(a) — slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$ (6 P)

Step 1 — means: $\bar{x} = \frac{\sum x_i}{n} = \frac{15}{5} = 3$, $\bar{y} = \frac{\sum y_i}{n} = \frac{55}{5} = 11$.

Step 2 — corrected cross-products:

$$S_{xy} = \sum x_i y_i - n\bar{x}\bar{y} = 195 - 5 \cdot 3 \cdot 11 = 195 - 165 = 30,$$

$$S_{xx} = \sum x_i^2 - n\bar{x}^2 = 55 - 5 \cdot 3^2 = 55 - 45 = 10.$$

Step 3 — slope and intercept:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{30}{10} = 3.000, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 11 - 3 \cdot 3 = 2.000,$$

giving the fitted line $\hat{y} = 2.000 + 3.000x$.

Interpretation. Slope: each extra weekly spot adds an expected 3,000 EUR in sales. Intercept: expected sales of 2,000 EUR at zero spots — a mild extrapolation ($x = 0$ unobserved; x ranges 1–5).

Problem 4 — solution (b): fitted values and residuals

Solution P4(b) — fitted values $\hat{y}_i$ and residuals e_i (4 P)

With $\hat{y}_i = 2 + 3x_i$ and $e_i = y_i - \hat{y}_i$; e.g. $\hat{y}_1 = 2 + 3 \cdot 1 = 5$ so $e_1 = 6 - 5 = 1$:

i	1	2	3	4	5
x_i	1	2	3	4	5
y_i	6	7	11	13	18
$\hat{y}_i = 2 + 3x_i$	5	8	11	14	17
$e_i = y_i - \hat{y}_i$	1	-1	0	-1	1

Check: $\sum_{i=1}^5 e_i = 1 - 1 + 0 - 1 + 1 = 0 \checkmark$

This is no coincidence: for any least-squares fit *with an intercept*, the residuals always sum to zero (one of the normal-equation identities).

Problem 4 — solution (c): RSS and RSE

Solution P4(c) — residual sum of squares and RSE (3 P)

Residual sum of squares (sum of squared residuals from part (b)):

$$\text{RSS} = \sum_{i=1}^5 e_i^2 = 1^2 + (-1)^2 + 0^2 + (-1)^2 + 1^2 = 1 + 1 + 0 + 1 + 1 = 4.000.$$

Residual standard error (with $n - 2 = 3$ degrees of freedom):

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{n-2}} = \sqrt{\frac{4}{3}} = \sqrt{1.333} = 1.155 \text{ (thousand EUR)}.$$

Interpretation. Even if the true regression line were known, observed weekly sales would still deviate from it by about 1.155 thousand EUR ($\approx 1,155$ EUR) on average. RSE is the estimate $\hat{\sigma}$ of σ , the standard deviation of the error ε .

Problem 4 — solution (d.1): standard error and t -test

Solution P4(d) — standard error and hypothesis test (part 1 of 2)

Hypotheses: $H_0: \beta_1 = 0$ vs. $H_1: \beta_1 \neq 0$, at level $\alpha = 0.05$.

Standard error of the slope:

$$SE(\hat{\beta}_1) = \frac{RSE}{\sqrt{S_{xx}}} = \frac{1.1547}{\sqrt{10}} = \frac{1.1547}{3.1623} = 0.365.$$

Test statistic ($df = n - 2 = 3$):

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} = \frac{3.000}{0.3651} = 8.216.$$

Decision: since $|t| = 8.216 > 3.182 = t_{0.975, 3}$, we *reject* H_0 at the 5% level — the slope is significantly different from zero.

Problem 4 — solution (d.2): confidence interval

Solution P4(d) — 95 % confidence interval for β_1 (part 2 of 2)

Using $\hat{\beta}_1 = 3.000$, $SE(\hat{\beta}_1) = 0.3651$ and $t_{0.975, 3} = 3.182$:

$$\begin{aligned}\hat{\beta}_1 \pm t_{0.975, 3} \cdot SE(\hat{\beta}_1) &= 3.000 \pm 3.182 \cdot 0.3651 \\ &= 3.000 \pm 1.162 \\ &= [1.838, 4.162].\end{aligned}$$

Zero is *not* contained in the interval — consistent with rejecting H_0 in part (d.1). **Conclusion:** there is a statistically significant *positive* linear association between the number of radio spots and weekly sales.

Test $\Leftrightarrow$ CI

A two-sided 5 % test rejects $H_0: \beta_1 = 0$ exactly when 0 falls outside the 95 % confidence interval.

Problem 4 — solution (e): TSS and R^2

Solution P4(e) — total sum of squares and R^2 (3 P)

Total sum of squares (deviations of y_i from $\bar{y} = 11$):

$$\text{TSS} = \sum_{i=1}^5 (y_i - \bar{y})^2 = (-5)^2 + (-4)^2 + 0^2 + 2^2 + 7^2 = 25 + 16 + 0 + 4 + 49 = 94.000.$$

Coefficient of determination (using $\text{RSS} = 4.000$):

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{4}{94} = \frac{90}{94} = 0.957.$$

Interpretation. About 95.7 % of the variance of weekly sales is explained by the linear regression on the number of radio spots. (In simple regression $R^2 = r^2$, so the correlation is $r = \sqrt{0.957} \approx 0.979$.)

Problem 4 — solution (f): prediction at $x = 6$

Solution P4(f) — prediction and its caveat (2 P)

Point prediction from $\hat{y} = 2 + 3x$ at $x = 6$:

$$\hat{y}(6) = 2 + 3 \cdot 6 = 2 + 18 = 20.000 \text{ thousand EUR} = 20,000 \text{ EUR.}$$

Caveat. $x = 6$ lies *outside* the observed range $x \in [1, 5]$ — this is an **extrapolation**. The linear relationship has not been observed there and may not hold; moreover the sample $n = 5$ is very small, so the estimates are themselves uncertain.

Extrapolation warning

A fitted line is only trustworthy *within* the range of the observed predictor values.

Problem 5 — task

Problem 5 (24 P) — Multiple regression output

$n = 60$ employees; salary (1,000 EUR/year); experience (years); female (dummy). OLS fit of salary = $\beta_0 + \beta_1 \text{experience} + \beta_2 \text{female} + \beta_3 (\text{experience} \times \text{female}) + \varepsilon$:

	coef	std err	t	$P > t $
Intercept	46.2000	1.100	42.000	<0.0001
experience	2.3500	0.250	9.400	<0.0001
female	-3.1000	1.550	-2.000	0.0504
experience:female	-0.8500	0.350	-2.429	0.0184

$R^2 = 0.712$; adj. $R^2 = 0.697$; $F = 46.15$ on 3 and 56 df; $\text{Prob}(F) < 0.0001$

(a) [6 P] Interpret all coefficients. (b) [4 P] Significance at 5 %; drop female? (c) [4 P] Fitted equation per group. (d) [4 P] Predictions at 10 years; gap. (e) [3 P] R^2 and F -test. (f) [3 P] U-shaped residual plot; VIF = 12.

Problem 5 — solution (a.1): intercept and experience

Solution P5(a) — baseline coefficients (part 1 of 2)

Salary is measured in 1,000 EUR per year; the *baseline* group is men (`female= 0`).

Intercept = 46.20. Expected annual salary of a *man* with 0 years of experience: 46,200 EUR.

experience = 2.35. For *men*, each additional year of experience raises the expected salary by 2,350 EUR. Because the model contains an interaction, this slope refers to the baseline group (`female= 0`) *only* — women have a different slope (see part a.2).

Reading a dummy–interaction model

With `female` and `experience:female` present, **Intercept** and **experience** describe the *men* only.

Problem 5 — solution (a.2): the dummy and the interaction

Solution P5(a) — gender difference and its slope (part 2 of 2)

$\text{female} = -3.10$. At 0 years of experience, women are expected to earn 3,100 EUR *less* than men — the difference between the two group *intercepts*.

$\text{experience:female} = -0.85$. The experience slope for women is 0.85 *lower* than for men:

$$\text{women's slope} = 2.35 - 0.85 = 1.50 \text{ thousand EUR per year.}$$

Equivalently, the expected salary gap between men and women *widens* by 850 EUR with every additional year of experience.

Problem 5 — solution (b): significance and the hierarchy principle

Solution P5(b) — significance and the hierarchy principle (4 P)

At $\alpha = 0.05$ compare each p -value with 0.05:

- Intercept, experience: $p < 0.0001 \Rightarrow$ *highly significant*.
- experience:female: $p = 0.0184 < 0.05 \Rightarrow$ *significant*. Consistency check:
 $t = \hat{\beta}_3 / \text{SE}(\hat{\beta}_3) = -0.85/0.35 = -2.429 \checkmark$ (matches the output).
- female: $p = 0.0504 > 0.05 \Rightarrow$ *not significant* at exactly the 5 % level (only marginally).

Should female be removed? No. By the **hierarchy principle**, if an interaction (experience×female) is kept, its main effects must be kept too — otherwise the interaction changes meaning. Besides, $p = 0.0504$ is borderline evidence, not evidence of *no* effect.

Problem 5 — solution (c): fitted equation per group

Solution P5(c) — separate lines for men and women (4 P)

Substitute $\text{female} = 0$ and $\text{female} = 1$ into $\widehat{\text{salary}} = 46.20 + 2.35 \text{exp} - 3.10 \text{female} - 0.85 (\text{exp} \times \text{female})$:

$$\text{Men (female} = 0\text{): } \widehat{\text{salary}} = 46.20 + 2.35 \text{experience}$$

$$\begin{aligned} \text{Women (female} = 1\text{): } \widehat{\text{salary}} &= (46.20 - 3.10) + (2.35 - 0.85) \text{experience} \\ &= 43.10 + 1.50 \text{experience} \end{aligned}$$

The two groups get *different intercepts and different slopes* — exactly what including the interaction term allows.

Problem 5 — solution (d): predictions and the gap at 10 years

Solution P5(d) — predicted salaries at 10 years (4 P)

Insert experience= 10 into the group equations from part (c):

$$\text{Man: } 46.20 + 2.35 \cdot 10 = 46.20 + 23.50 = 69.70 \Rightarrow 69,700 \text{ EUR}$$

$$\text{Woman: } 43.10 + 1.50 \cdot 10 = 43.10 + 15.00 = 58.10 \Rightarrow 58,100 \text{ EUR}$$

Estimated gap (woman – man): $58.10 - 69.70 = -11.60$ thousand EUR — women earn 11,600 EUR less.
In terms of the coefficients the gap at experience = 10 is

$$\hat{\beta}_2 + 10 \hat{\beta}_3 = -3.10 - 8.50 = -11.60 \text{ thousand EUR.}$$

Problem 5 — solution (e): R^2 and the F -test

Solution P5(e) — overall fit and joint significance (3 P)

$R^2 = 0.712$. The three model terms (experience, female and their interaction) jointly explain 71.2 % of the variance of salaries in the sample.

F -test — is the model useful at all?

$$H_0: \beta_1 = \beta_2 = \beta_3 = 0 \quad \text{vs.} \quad H_1: \text{at least one } \beta_j \neq 0.$$

The output gives $F = 46.15$ on 3 and 56 degrees of freedom with $p < 0.0001 \Rightarrow \text{reject } H_0$: at least one predictor is related to salary, so the model is useful overall.

Problem 5 — solution (f.1): non-linearity

Solution P5(f)(i) — U-shaped residual plot (part 1 of 2)

A residual-vs-fitted plot with a clear **U-shape** (positive residuals at small and large fitted values, negative in between) signals **non-linearity**: the straight-line model misses systematic *curvature*, so the residuals are not patternless as they should be.

Remedies.

- Add a non-linear term, e.g. experience^2 (polynomial regression);
- Transform variables, e.g. model $\log(\text{salary})$ or use $\sqrt{X}$.

Diagnostic rule

A good linear fit shows a *structureless* residual cloud; any systematic shape points to a violated assumption.

Problem 5 — solution (f.2): multicollinearity (VIF)

Solution P5(f)(ii) — what $VIF = 12$ means (part 2 of 2)

Definition. $VIF_j = \frac{1}{1 - R_{X_j|X_{-j}}^2}$, where $R_{X_j|X_{-j}}^2$ comes from regressing X_j on all *other* predictors. Solving $VIF_{\text{age}} = 12$:

$$R_{\text{age}|\text{rest}}^2 = 1 - \frac{1}{12} = 0.917.$$

So age is almost a linear combination of the other predictors — strong **multicollinearity** (common rules of thumb flag $VIF > 5$ or > 10).

Consequence. $SE(\hat{\beta}_{\text{age}})$ is inflated by a factor $\sqrt{12} \approx 3.5$: the coefficient is estimated very imprecisely, its sign may flip between samples, and its t -test loses power — even though the model's overall fit and predictions may be unaffected.